

Airline Analytics

Comprehensive EDA of Airline Performance, Flight Behaviour, Engagement & Revenue Insights using Powerbi and Python

1. Project Overview and Objective

1. Analyze airline performance based on factors such as customer loyalty, airlines customer base, customer demo graphics, flight behaviour, spending patterns, and loyalty engagement and passenger-related information available in the dataset. Perform Exploratory Data Analysis (EDA) using Pandas, NumPy, Matplotlib, and Seaborn to understand important patterns and relationships.
2. Clean and preprocess the data by handling missing values, duplicate records, incorrect data types, and inconsistent values
3. Generate meaningful business insights from the analysis that can support better decision-making and improve airline operations.
4. Visualize key insights using appropriate charts such as bar charts, line charts, histograms, box plots, count plots, and heatmaps.
5. Python data-analysis skills including data cleaning, filtering, grouping, aggregation, statistical analysis, and visualization.

2. Data Sources

Domain

Domain: Aviation & Transportation Analytics, with a strong focus on Data Analytics and Business Intelligence.

Dataset Information

****Source:**** kaagle.com <https://www.kaggle.com/datasets/mahoora00135/flights>

This project analyzes airline customer and flight-activity data using Python for data cleaning, exploratory data analysis, statistical investigation and features, followed by Power BI for interactive business reporting. The analysis focuses on customer loyalty,

flight frequency, travel distance, revenue, loyalty points, discount behaviour and customer engagement.

The dataset contains 2,499 customer records and 23 columns, including customer attributes, loyalty tier, flight activity, spending, distance travelled, points and date-related fields. The Power BI solution is designed as two complementary dashboards: (1) Airline Performance & Customer Loyalty and (2) Flight Behaviour, Engagement & Revenue Insights.

3. Problem Statement

The Airline Analytics project provides a clear and comprehensive understanding of airline and flight-related data through Python-based analysis. By applying data cleaning, exploratory data analysis, statistical techniques, and data visualization, the project uncovers important patterns, trends, relationships, and hidden insights within the dataset.

The analysis helps identify airline performance, high probability of inactive, based on travel frequency recency and other key factors that influence airline operations. Visualizations make complex data easier to interpret, while evidence-based findings help highlight areas of strong performance and potential improvement. Overall, the project transforms raw airline data into meaningful and actionable insights, supporting data-driven decision-making and better understanding of airline operations

Descriptive Analysis:

Customer loyalty analysis

- *Customer segmentation

- *Customer spending analysis

- *Flight behavior analysis

- *Customer inactivity / churn-risk analysis

- *Discount analysis

- *Customer lifetime / relationship analysis

The analysis identifies high-value and loyal customers based on flight frequency, spending, distance travelled, loyalty tier, and points accumulated. It also identifies inactive or at-risk customers using their recency of travel and engagement behavior. These insights can help the airline improve customer retention, personalize loyalty programs, target promotional campaigns, and focus resources on customers with the highest revenue potential.

Predictive Analysis

Customer Churn Prediction:

- *long time since last flight

- *low flight frequency

- *low spending

- *long intervals between flights

- *low loyalty tier

- *have a higher probability of becoming inactive.

- *Target high-risk customers with personalized offers, bonus points, discounted flights, or loyalty rewards to encourage them to fly again.

Predict customer loyalty:

The predictive model identifies customers who have a high probability of becoming inactive based on their travel frequency, recency, spending, distance travelled, loyalty status, and engagement behavior. These predictions allow the airline to proactively target high-risk customers with personalized retention campaigns, helping reduce customer churn and protect future revenue.

4. Attribute (Column /Features) Details

Attribute Name	Data Type	Description
MEMBER_NO	Integer Numeric	/ Unique membership number assigned to each airline customer.
FFP_DATE	Date	Date when the customer joined the Frequent Flyer Program (FFP).
FIRST_FLIGHT_DATE	Date	Date of the customer's first recorded flight.
GENDER	Categorical Text	/ Gender of the customer.
FFP_TIER	Integer Categorical	/ Frequent Flyer Program membership level or loyalty tier.
WORK_CITY	Categorical Text	/ City where the customer works.
WORK_PROVINCE	Categorical Text	/ Province or state where the customer works.
WORK_COUNTRY	Categorical Text	/ Country where the customer works.
AGE	Integer Numeric	/ Age of the customer in years.
LOAD_TIME	Date / DateTime	Date or time representing the data observation or analysis period.
FLIGHT_COUNT	Integer Numeric	/ Total number of flights taken by the customer.
BP_SUM	Numeric	Total accumulated points associated with flight bookings or benefits.

Attribute Name	Data Type	Description
MEMBER_NO	Integer Numeric	/ Unique membership number assigned to each airline customer.
SUM_YR_1	Numeric Decimal	/ Customer's total flight-related expenditure or revenue during Year 1.
SUM_YR_2	Numeric Decimal	/ Customer's total flight-related expenditure or revenue during Year 2.
SEG_KM_SUM	Numeric Decimal	/ Total distance travelled by the customer, measured in kilometers.
LAST_FLIGHT_DATE	Date	Date of the customer's most recent flight.
LAST_TO_END	Integer Numeric	/ Number of days since the customer's last flight.
AVG_INTERVAL	Numeric Decimal	/ Average number of days between the customer's flights.
MAX_INTERVAL	Integer Numeric	/ Maximum number of days between two consecutive flights.
EXCHANGE_COUNT	Integer Numeric	/ Number of times the customer exchanged or redeemed points.
avg_discount	Numeric Decimal	/ Average discount received by the customer on flights or purchases.
Points_Sum	Numeric Decimal	/ Total points accumulated by the customer.
Point_NotFlight	Integer Numeric	/ Number of points earned through non-flight activities.

5. Tools & Technologies

Tool / Technology	Purpose in the Project
Python	Used for data cleaning, exploratory data analysis (EDA), statistical analysis, and data visualization.
Google Colab	Used as the development environment for writing and executing Python code.
Pandas	Used for data loading, data cleaning, handling missing values, filtering, transformation, and data manipulation.
NumPy	Used for numerical calculations, mathematical operations, and data processing.
Matplotlib	Used to create charts and visualizations such as line charts, bar charts, histograms, and scatter plots.
Seaborn	Used to create statistical visualizations such as count plots, box plots, pair plots, and heatmaps.
Power BI	Used to create interactive dashboards and present airline customer, flight, loyalty, spending, and travel insights.
DAX (Data Analysis Expressions)	Used to create calculated columns and measures for KPIs, customer metrics, spending, flight analysis, and other business calculations.
Power BI Data Model	Used to organize and establish relationships between datasets and support interactive analysis
CSV / Excel	Used as the source data format for storing and importing the airline customer dataset.

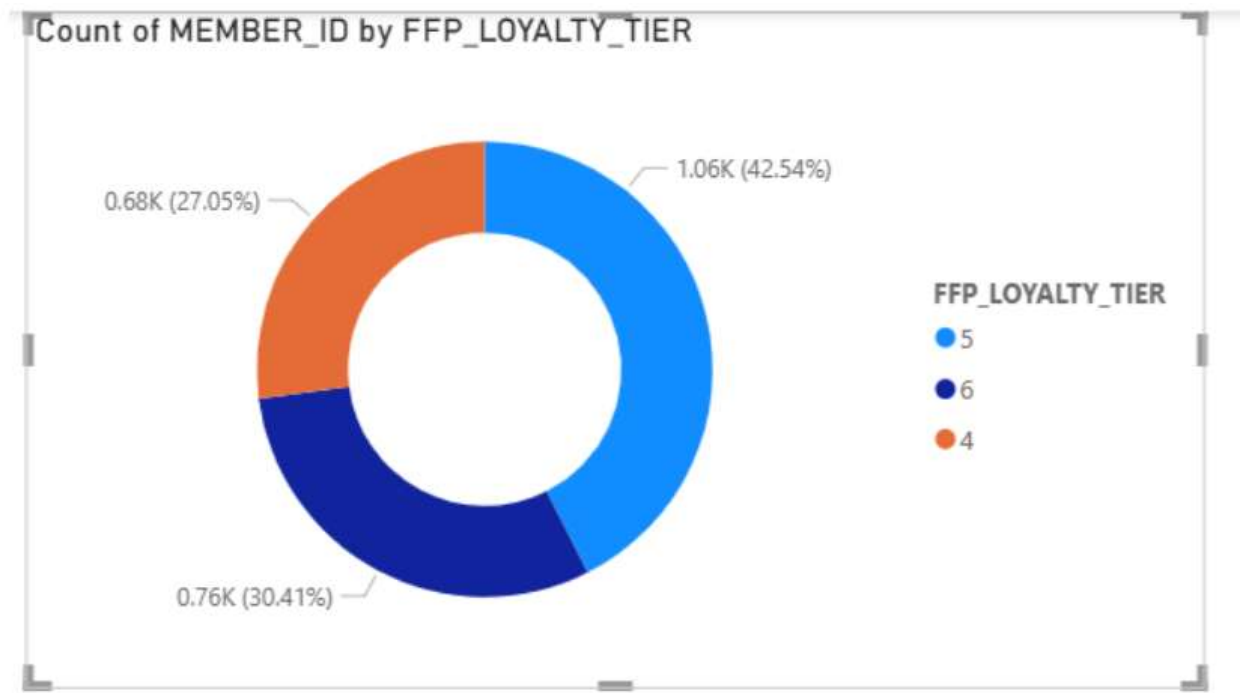
6. Data Pre-Processing

- Data Pre-Processing: The airline customer dataset was preprocessed using Python before performing exploratory and business analysis
- The preprocessing included data loading, dataset inspection, missing-value analysis, data-type conversion, duplicate checking, outlier and skewness analysis, and feature preparation.

The required Python libraries were imported for data manipulation, numerical analysis, and visualization.

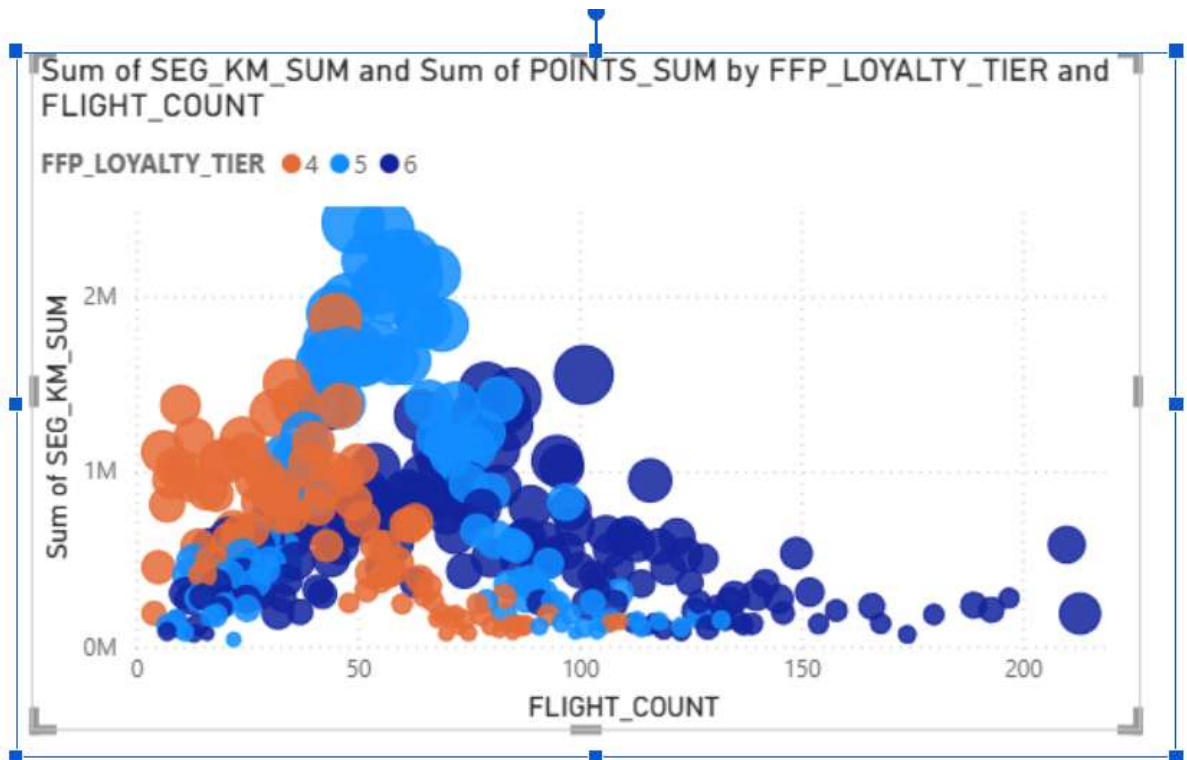
- **Pandas** – Data loading and manipulation
- **NumPy** – Numerical operations
- **Matplotlib** – Data visualization
- **Seaborn** – Statistical visualization.
- Date columns were converted into appropriate datetime formats, while numerical and categorical variables were standardized for analysis. Highly skewed numerical variables were transformed using logarithmic transformation where appropriate.
- The final cleaned dataset was validated and prepared for EDA, customer segmentation, loyalty analysis, and interactive dashboard development in Power BI.

Data visualisations in Power BI:



Donut chart

The loyalty-tier distribution shows the strength of the airline's customer loyalty program. FFP-loyalty_tier 5 has the largest customer memberships groups.



Scatter chart

The customers who travelled longer distances with FFP_loyalty_tier 4 have taken less flights.

Membership points with 6 have travelled minimal distances but taken more flights.

8. Analysis and Visualizations (Power BI)

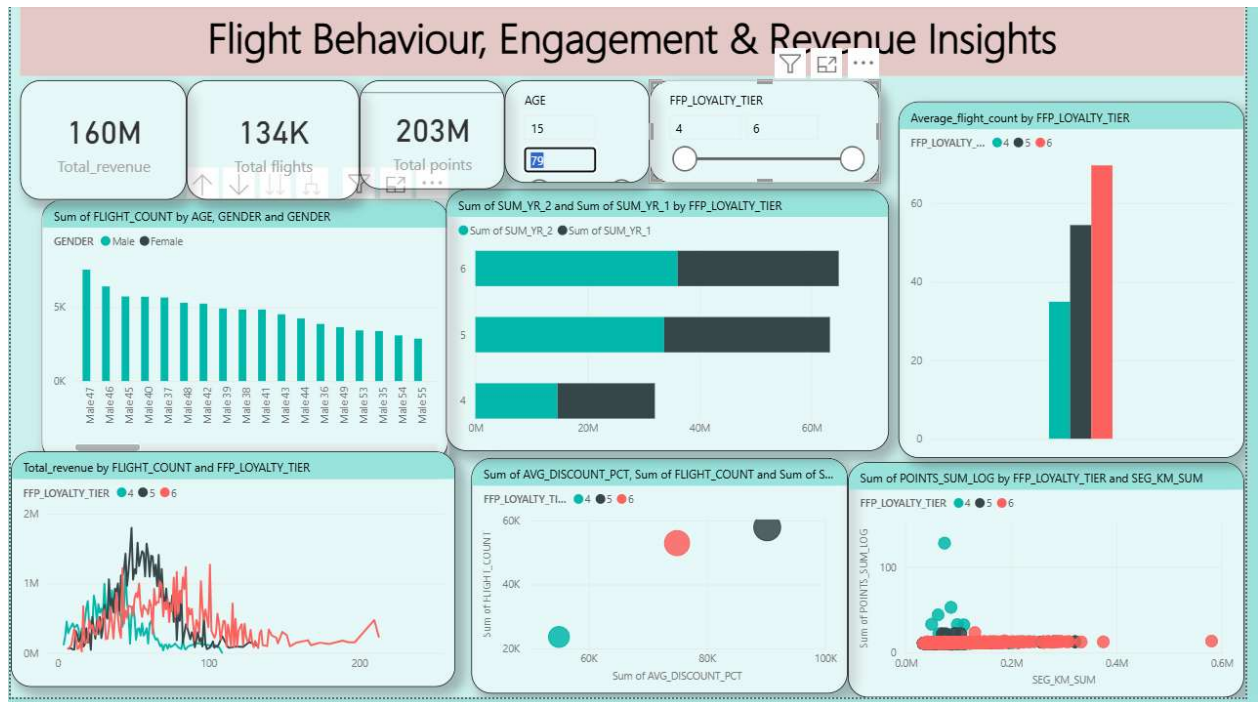
Dashboard 1



“The first dashboard focuses on overall airline performance and customer loyalty. It helps us understand who the customers are, how they are distributed across loyalty tiers, and how their travel activity contributes to airline performance.

From the analysis, Tier 5 represents the largest customer segment. We can also see that customers with higher flight frequency tend to travel greater distances and contribute more to the airline's overall business value.

Dashboard 2



9. Insights & Conclusions

The dashboard analyses customer flight behaviour and engagement.

Average flight count helps measure how frequently customers travel.

Flight count verses revenue helps identify customers who generate higher business value through frequent travel.

Flight count versus distance shows the relationship between travel frequency and total distance.

Discount analysis helps understand whether customers receiving different discount levels show different flight activity.

Points analysis helps identify customers who are highly engaged with the airline's loyalty program.